

NAME : Leena K

ROLL NO : CH.SC.U4CSE24122

WEEK - 5

QUICK SORT:

Logic:

classmate
Date _____
Page _____

By first element

157	110	147	122	111	149	151	141	123	112	117	133
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot element = 157

110	147	122	111	149	151	141	123	112	117	133	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 110

110	147	122	111	149	151	141	123	112	117	133	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 147

110	122	111	141	123	112	117	133	147	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 122

110	111	112	117	122	123	133	141	147	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 111

110	111	112	117	122	123	133	141	147	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

By last element

157	110	147	122	111	149	151	141	123	112	117	133
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

S1:

110	122	111	123	112	117	133	157	147	149	151	141
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 117 pivot = 141

S2:

110	111	112	117	122	123	133	141	157	147	149	151
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 112 pivot = 123 pivot = 151

S3:

110	111	112	117	122	123	133	141	147	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

By random element:

S1:

157	110	147	122	111	149	151	141	123	112	117	133
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 123

S2:

110	122	111	112	117	123	157	147	149	151	141	133
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 151

S3:

110	111	122	112	117	123	147	149	141	133	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 112

pivot = 149

S4:

110	111	112	122	117	123	147	141	133	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

pivot = 122

pivot = 149

S5:

110	111	112	117	122	123	133	141	147	149	151	157
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

First Element as Pivot

Code :

```
1 //CH.SC.U4CSE24122
2 #include <stdio.h>
3 int partition(int arr[], int low, int high) {
4     int pivot = arr[low];
5     int i = low + 1;
6     int j = high;
7     int temp;
8     while (i <= j) {
9         while (i <= j && arr[i] <= pivot)
10             i++;
11         while (arr[j] > pivot)
12             j--;
13         if (i < j) {
14             temp = arr[i];
15             arr[i] = arr[j];
16             arr[j] = temp;
17         }
18     }
19     arr[low] = arr[j];
20     arr[j] = pivot;
21     return j;
22 }
23 void quickSort(int arr[], int low, int high) {
24     if (low < high) {
25         int p = partition(arr, low, high);
26         quickSort(arr, low, p - 1);
27         quickSort(arr, p + 1, high);
28     }
29 }
30 int main(){
31     printf("CH.SC.U4CSE24122\n");
32     int arr[] = {157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133};
33     int n = 12;
34     int i;
35     quickSort(arr, 0, n - 1);
36     printf("Sorted array:\n");
37     for (i = 0; i < n; i++)
38         printf("%d ", arr[i]);
39     return 0;
40 }
41
```

Output :

```
CH.SC.U4CSE24122
Sorted array:
110 111 112 117 122 123 133 141 147 149 151 157
-----
Process exited after 0.1208 seconds with return value 0
Press any key to continue . . .
```

Last Element as Pivot

Code :

```
1 //CH.SC.U4CSE24122
2 #include <stdio.h>
3 int partition(int arr[], int low, int high) {
4     int pivot = arr[high];
5     int i = low - 1;
6     int temp, j;
7     for (j = low; j < high; j++) {
8         if (arr[j] <= pivot) {
9             i++;
10            temp = arr[i];
11            arr[i] = arr[j];
12            arr[j] = temp;
13        }
14    }
15    temp = arr[i + 1];
16    arr[i + 1] = arr[high];
17    arr[high] = temp;
18    return i + 1;
19 }
20 void quickSort(int arr[], int low, int high) {
21     if (low < high) {
22         int p = partition(arr, low, high);
23         quickSort(arr, low, p - 1);
24         quickSort(arr, p + 1, high);
25     }
26 }
27 int main() {
28     printf("CH.SC.U4CSE24122\n");
29     int arr[] = {157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133};
30     int n = 12;
31     int i;
32     quickSort(arr, 0, n - 1);
33     printf("Sorted array:\n");
34     for (i = 0; i < n; i++)
35         printf("%d ", arr[i]);
36     return 0;
37 }
38
```

Output :

```
CH.SC.U4CSE24122
Sorted array:
110 111 112 117 122 123 133 141 147 149 151 157
-----
Process exited after 0.1193 seconds with return value 0
Press any key to continue . . .
```

Random Element as Pivot

Code :

```
1 //CH.SC.U4CSE24122
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <time.h>
5 int partition(int arr[], int low, int high) {
6     int pivot = arr[high];
7     int i = low - 1;
8     int temp;
9     for (int j = low; j < high; j++) {
10         if (arr[j] <= pivot) {
11             i++;
12             temp = arr[i];
13             arr[i] = arr[j];
14             arr[j] = temp;
15         }
16     }
17     temp = arr[i + 1];
18     arr[i + 1] = arr[high];
19     arr[high] = temp;
20     return i + 1;
21 }
22 int randomPartition(int arr[], int low, int high) {
23     int r = low + rand() % (high - low + 1);
24     int temp = arr[r];
25     arr[r] = arr[high];
26     arr[high] = temp;
27
28     return partition(arr, low, high);
29 }
30 void quickSort(int arr[], int low, int high) {
31     if (low < high) {
32         int p = randomPartition(arr, low, high);
33         quickSort(arr, low, p - 1);
34         quickSort(arr, p + 1, high);
35     }
36 }
37 int main() {
38     printf("CH.SC.U4CSE24122\n");
39     int arr[] = {157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133};
40     int n = 12;
41     int i;
42     srand(time(0));
43     quickSort(arr, 0, n - 1);
44     printf("Sorted array:\n");
45     for (i = 0; i < n; i++)
46         printf("%d ", arr[i]);
47     return 0;
48 }
49
```

Output :

```
CH.SC.U4CSE24122
Sorted array:
110 111 112 117 122 123 133 141 147 149 151 157
-----
Process exited after 0.1714 seconds with return value 0
Press any key to continue . . .
```

Conclusion:

Based on the array order, position of the pivot element changes.

So, the efficiency doesn't depend on the position of the pivot element changes